

Show all your work in the space provided for each problem. All problems will be graded for correctness, completeness, conciseness, and clarity. Do your best. Don't cut corners.

1. (10 pts) Use math induction to prove that $\sum_{i=1}^n 2i = n^2 + n$ for all positive integers n .

Let $P(n)$ be the statement $\sum_{i=1}^n 2i = n^2 + n$.

Base case: $P(1)$ is true since $\sum_{i=1}^1 2i = 2 = 1^2 + 1$ ✓

Induction Step: If $\sum_{i=1}^k 2i = k^2 + k$ then $\sum_{i=1}^{k+1} 2i = (k+1)^2 + (k+1)$

$$\sum_{i=1}^{k+1} 2i = \sum_{i=1}^k 2i + 2(k+1) = k^2 + k + 2k + 2 = (k^2 + 2k + 1) + (k+1) = (k+1)^2 + (k+1)$$

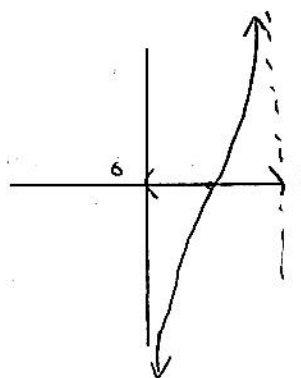
Conclusion: Since $P(1)$ is true and $P(k+1)$ is true whenever $P(k)$ is true we know by weak math induction that $P(n)$ is true for all $n \geq 1$

2. (6 pts) State carefully the difference between the induction hypothesis in weak induction and the induction hypothesis in strong induction?

Weak: The statement $P(k)$ is true [show $P(k+1)$ is true.]

Strong: The statements $P(a)$ through $P(k)$ are true [show that $P(k+1)$ is true]

3. (4 pts) Draw a graph of a function that shows a one-to-one correspondence between the open interval $(0,1)$ and $\mathbb{R}$.



$$\frac{2x-1}{x(1-x)}$$

4. (12 pts) Let f be a function defined on $\mathbb{R}$.

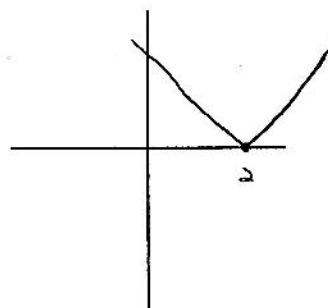
a) What limit must be satisfied for f to be continuous at $x = c$?

$$\lim_{x \rightarrow c} f(x) = f(c)$$

b) What limit must exist for f to be differentiable at $x = c$?

$$\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} \text{ must be a number}$$

c) Draw the graph of a function that is continuous at $x = 2$ but clearly is not differentiable at $x = 2$.



5. (12 pts) Let $f: \mathbb{Z} \rightarrow \mathbb{Z}$ be defined by $f(x) = 2x$.

a) Prove or disprove that f is one-to-one.

f is 1-1. If $f(a) = f(b)$ we have $2a = 2b$.
Dividing through by 2 we get $a = b$.
so f is 1-1.

b) Prove or disprove that f maps onto $\mathbb{Z}$.

f is not onto $\mathbb{Z}$ since there is no $x \in \mathbb{Z}$ such
that $f(x) = 1$.

$2x = 1$ forces $x = \frac{1}{2}$ which is not in $\mathbb{Z}$.

6. (9 pts) Let $A = \{1, 2, 3, 4\}$ and $B = \{a, b, c\}$

a) List a set of ordered pairs that will define a function with domain A and codomain B that is not a surjection (onto).

$\{(1, a), (2, a), (3, a), (4, a)\}$

b) List a set of ordered pairs that will define a function with domain B and codomain A that is one-to-one.

$\{(a, 1), (b, 2), (c, 3)\}$

c) Explain why there is no function with domain A and codomain B that is one-to-one?

Any function with domain A must have exactly 4 ordered pairs. But there are only 3 possible function values in B so at least one function value in B must be repeated.

7. (12 pts) Let f be a mapping from domain X to codomain Y .

a) Write an if-then statement that defines what it means for f to be a function.

$$\text{If } a = b \text{ then } f(a) = f(b).$$

b) Write an if-then statement that defines what it means for f to be one-to-one.

$$\text{If } f(a) = f(b) \text{ then } a = b$$

8. (10 pts) Use the **definition** of derivative to find $f'(4)$ if $f(x) = x^2 + x$.

$$\begin{aligned} f'(4) &= \lim_{x \rightarrow 4} \frac{f(x) - f(4)}{x - 4} = \lim_{x \rightarrow 4} \frac{(x^2 + x) - 20}{x - 4} \\ &= \lim_{x \rightarrow 4} \frac{(x - 4)(x + 5)}{x - 4} = \lim_{x \rightarrow 4} x + 5 = 9 \end{aligned}$$

9. (6 pts) Let S be the set $\{1, 2, 3, 4\}$ and let f be the function mapping S to S defined by the following set of ordered pairs – $\{(1,4), (2,2), (3,1), (4,3)\}$

a) What is $f(1)$? $f(1) = 4$

b) What is $f^{-1}(1)$? $f^{-1}(1) = 3$

c) What is $(f \circ f)(1)$ $f(f(1)) = f(4) = 3$

10. (9 pts) Let $h(x) = x + f(x)g(x)$.

a) Using known derivative rules we get

$$h'(x) = 1 + f(x) \cdot g'(x) + g(x) \cdot f'(x)$$

b) Using the values given in the table

	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
$x=5$	1	2	3	4

we get $h(5) = 5 + f(5) \cdot g(5) = 5 + 1 \cdot 3 = 8$

and $h'(5) = 1 + f(5) \cdot g'(5) + g(5) \cdot f'(5) = 1 + 1 \cdot 4 + 3 \cdot 2 = 11$

11. (10 pts) Use Cauchy's definition to prove that $\lim_{x \rightarrow 3} 2x^2 - 8 = 10$

Let $\varepsilon > 0$ be any positive real number

Begin by assuming $|x-3| \leq 1$ so $2 \leq x \leq 4$

$$\begin{aligned} \text{In this case } |f(x) - L| &= |2x^2 - 18| = |2(x^2 - 9)| = |2(x+3)(x-3)| \\ &= |2x+6| \cdot |x-3| \end{aligned}$$

The max value of $|2x+6|$ when $2 \leq x \leq 4$ will be 14.

So if $|x-3| \leq 1$ we know $|f(x) - L| \leq 14 \cdot |x-3| < \varepsilon$ if $|x-3| < \frac{\varepsilon}{14}$

Choose $\delta = \min \left\{ 1, \frac{\varepsilon}{14} \right\}$.

Therefore if $0 < |x-3| < \delta$ we have $|x-3| < 1$ and $|x-3| < \frac{\varepsilon}{14}$.

This implies $|f(x) - L| = |2x+6| \cdot |x-3| < 14 \cdot \frac{\varepsilon}{14} = \varepsilon$

Thus $\lim_{x \rightarrow 3} 2x^2 - 8 = 10$